OLAP Operations Demonstration

Project: University Library Analytics Data Warehouse
Database: MySQL (`library_dw`)

Introduction

OLAP (Online Analytical Processing) allows users to interactively analyze multi-dimensional data from different perspectives. Below are the implementations of the required OLAP operations using our Star Schema.

1. Drill-Down Analysis

Definition: Navigating from a summarized data level to a more detailed level (e.g., Year -> Quarter -> Month).

Business Question: What is the total number of digital downloads in 2024, broken down by quarter and month?

SQL Query:

```
SELECT
    d.year,
    d.quarter,
    d.month,
    SUM(f.download_count) AS total_downloads
FROM library_usage_fact f
JOIN dim_date d ON f.date_key = d.date_key
WHERE d.year = 2024
GROUP BY d.year, d.quarter, d.month WITH ROLLUP;
```

Note: The `WITH ROLLUP` operator automatically generates subtotals for each quarter and a grand total for the year.

2. Roll-Up Analysis

Definition: Aggregating detailed data into a higher-level summary (e.g., Daily -> Monthly).

Business Question: What are the total room booking hours aggregated by month?

SQL Query:

```
SELECT
    d.year,
    d.month,
    SUM(f.duration_hours) AS total_room_hours
FROM library_usage_fact f
JOIN dim_date d ON f.date_key = d.date_key
WHERE f.duration_hours > 0
GROUP BY d.year, d.month
ORDER BY d.year, d.month;
```

3. Slicing

Definition: Selecting a single dimension to view a specific subset of the data.

Business Question: View all library usage events that occurred only in Q1 of 2024.

SQL Query:

```
SELECT
    d.year,
    d.quarter,
    SUM(f.download_count) AS digital_downloads,
    SUM(f.loan_days) AS physical_loan_days
FROM library_usage_fact f
JOIN dim_date d ON f.date_key = d.date_key
WHERE d.year = 2024 AND d.quarter = 1
GROUP BY d.year, d.quarter;
```

4. Dicing

Definition: Selecting a sub-cube by filtering on two or more dimensions simultaneously.

Business Question: Analyze the digital downloads for Engineering students specifically during the month of March 2024.

SQL Query:

```
SELECT
    s.department,
    d.month,
    SUM(f.download_count) AS engineering_downloads
FROM library_usage_fact f
JOIN dim_date d ON f.date_key = d.date_key
JOIN dim_student s ON f.student_key = s.student_key
WHERE s.department = 'Engineering'
  AND d.year = 2024
  AND d.month = 3
GROUP BY s.department, d.month;
```

5. Pivot Analysis

Definition: Rotating data to view it in a matrix format, comparing two dimensions against each other.

Business Question: Compare the total resource usage (events) across all departments, split by the type of resource used (Physical Books vs. E-books vs. Journals).

SQL Query:

```
SELECT
    s.department,
    SUM(CASE WHEN r.resource_type = 'Physical Book' THEN 1 ELSE 0 END) AS physical_books,
    SUM(CASE WHEN r.resource_type = 'E-book' THEN 1 ELSE 0 END) AS e_books,
    SUM(CASE WHEN r.resource_type = 'Journal' THEN 1 ELSE 0 END) AS journals
FROM library_usage_fact f
JOIN dim_student s ON f.student_key = s.student_key
JOIN dim_resource r ON f.resource_key = r.resource_key
WHERE s.department IS NOT NULL
GROUP BY s.department
ORDER BY s.department;
```